

4-Vesta, Orion Nebula, Andromeda Galaxy

The background of the slide is a deep space image. In the center, there is a bright, glowing yellowish-white core, which is the central region of the Andromeda Galaxy. Surrounding this core are faint, wispy structures of gas and dust, characteristic of nebulae. The entire scene is set against a dark, star-filled background with many small, distant stars visible.

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Asteroids

- Near-Earth Asteroids are Easily Accessible
- There are over 16,000 near-Earth asteroids that share a similar orbit to Earth.
- Detecting asteroids and labeling It's threat level to Earth.
- NEA – Near Earth Asteroid: Labeled if closer than 1.3 AU (120,800,000 mi)
- PHA – Potentially Hazardous Asteroid: labeled if asteroid is closer than .05 AU (4,650,000 mi)
- Rich in Resources
- There are an estimated two trillion tons of water available on near-Earth asteroids. This water can be used to sustain human life and as propellant for spacecraft.

4-Vesta

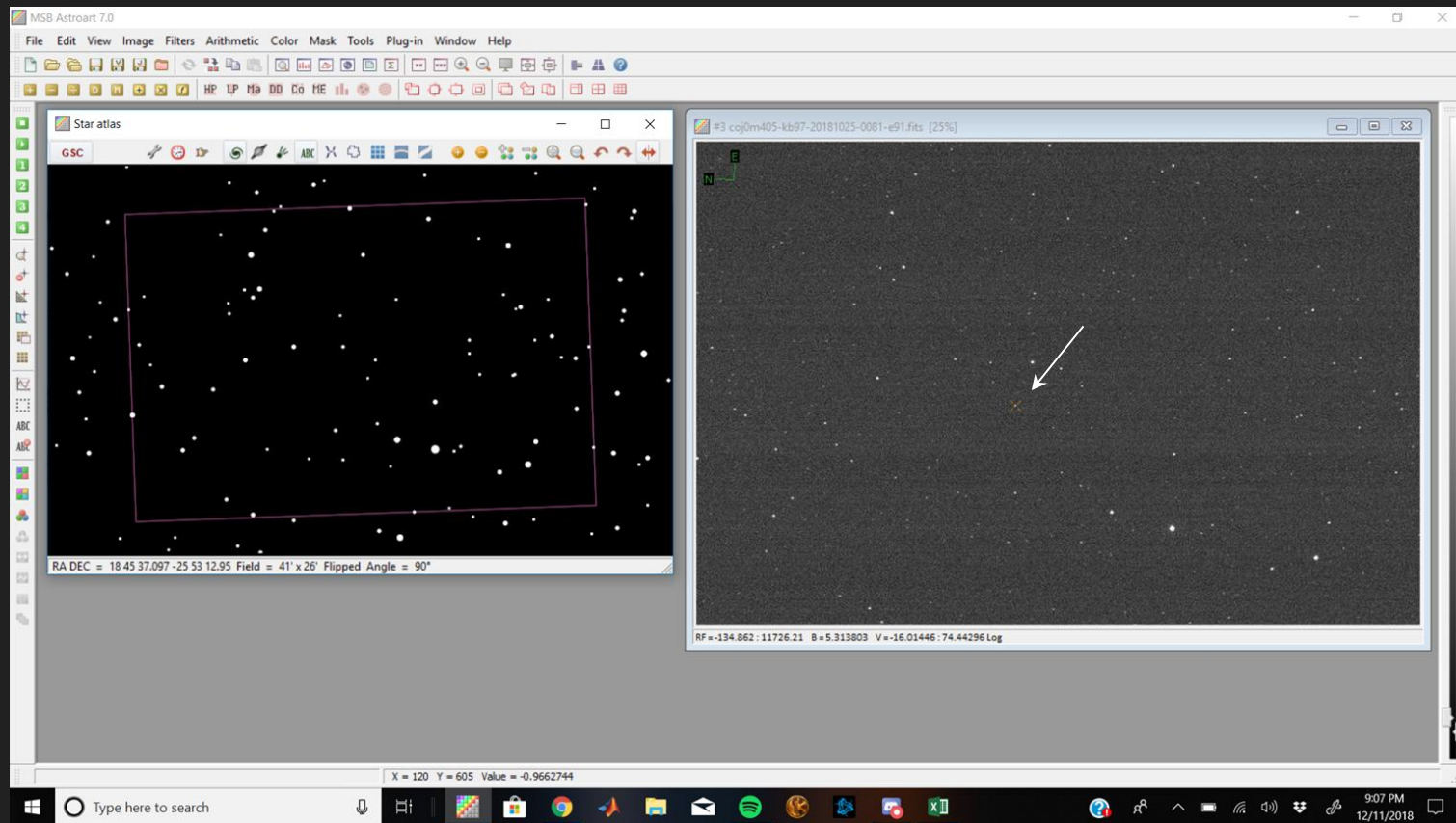
- Discovered by the German astronomer Heinrich Wilhelm Olbers on 29 March 1807
 - Mean diameter of 525 kilometres
 - It is one of the largest and most massive asteroids in the asteroid belt, second only to Ceres (a dwarf planet)
- it contributes an estimated 9% of the mass of the asteroid



Observation

Target Name	RA	Dec	Filter	# Exposures	Integration Time (s)
4 Vesta	281.2875	-25.7169	Clear	10	.5
4 Vesta	281.2875	-25.7169	SDSS u'	3	1
4 Vesta	281.2875	-25.7169	SDSS i'	3	1
4 Vesta	281.2875	-25.7169	SDSS g'	3	.5
4 Vesta	281.2875	-25.7169	SDSS r'	3	1

Locating 4-Vesta



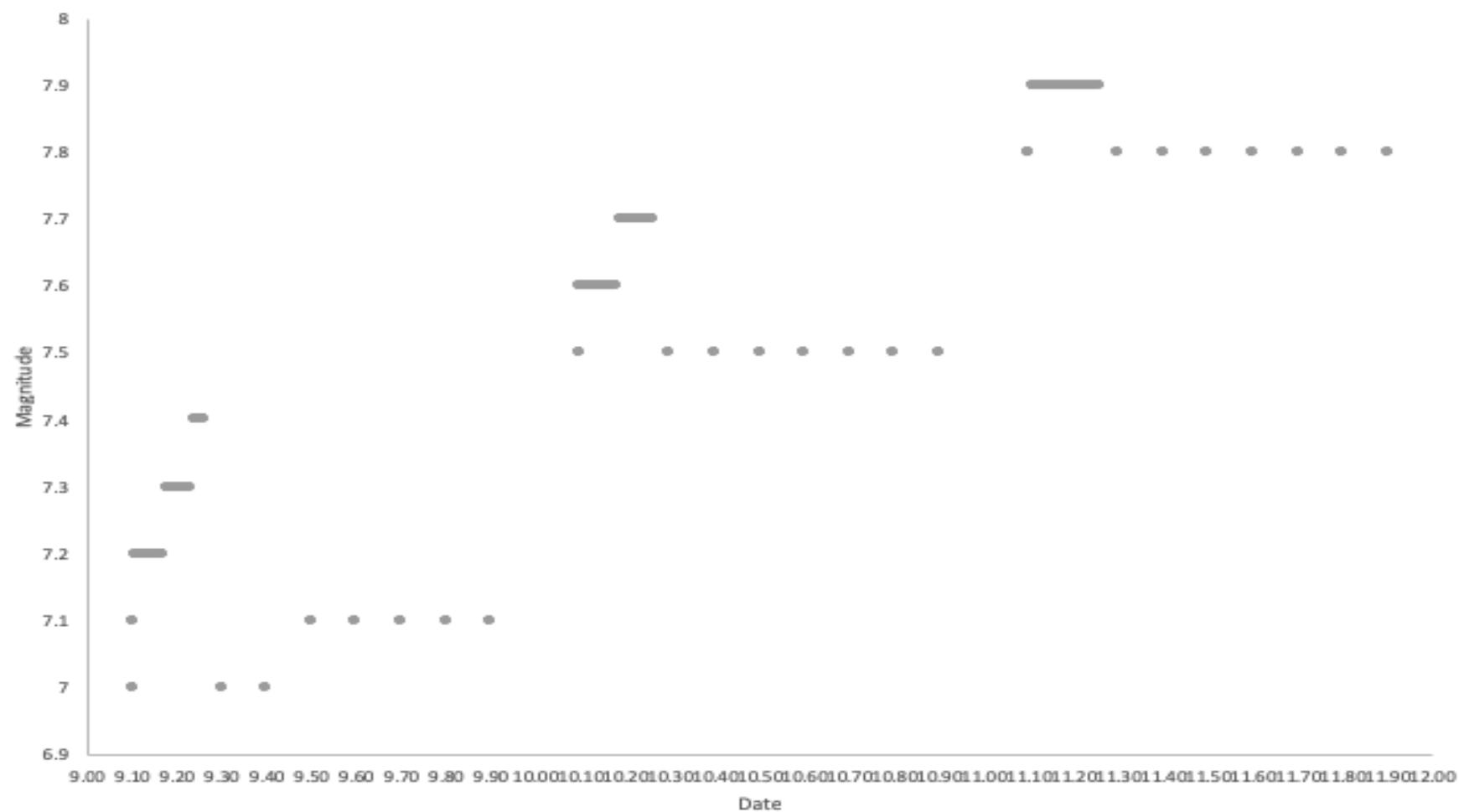
Light Curves

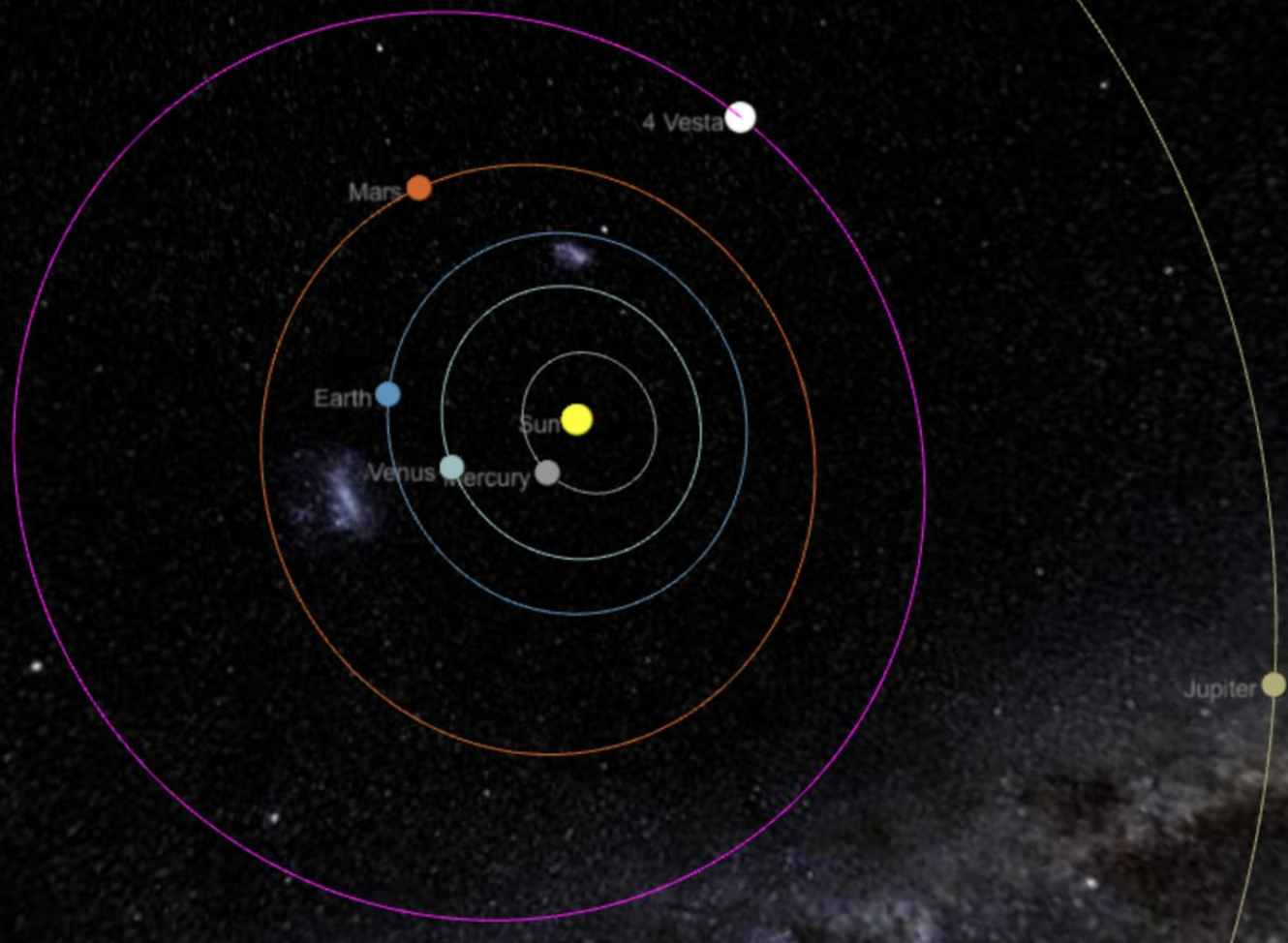
- One of the things we hoped to learn through observation of near-Earth objects is their exact rotation rate, by taking a series of observations of the object over time, and plotting the change in brightness.
- Unfortunately we did not know the extent of data needed to produce accurate light curves.
- Instead of using our data we pulled some secondary source data from <https://theskylive.com/vesta-info> and used that to plot our own light curve
- The three plots we have made are magnitude vs phase, magnitude vs date, and altered magnitude vs phase.

Data from TheSkyLive

Year	Month	Date	RA	Dec	Magnitude	Filter	Period (hours)	Rotation(days_	Phase	Altered Mag
Enter 8	11	11.1	19 ^h 12 ^m 22 ^s	-25°23'05"	7.8	V	5.342	0.215	0.627906977	7.536
2018	11	11.3	19 ^h 14 ^m 14 ^s	-25°20'55"	7.8	V	5.342	0.215	0.558139535	7.536
2018	11	11.4	19 ^h 16 ^m 07 ^s	-25°18'40"	7.8	V	5.342	0.215	0.023255814	7.536
2018	11	11.5	19 ^h 18 ^m 01 ^s	-25°16'18"	7.8	V	5.342	0.215	0.488372093	7.536
2018	11	11.6	19 ^h 19 ^m 54 ^s	-25°13'49"	7.8	V	5.342	0.215	0.953488372	7.536
2018	11	11.7	19 ^h 21 ^m 49 ^s	-25°11'14"	7.8	V	5.342	0.215	0.418604651	7.536
2018	11	11.8	19 ^h 23 ^m 43 ^s	-25°08'33"	7.8	V	5.342	0.215	0.88372093	7.536
2018	11	11.9	19 ^h 25 ^m 38 ^s	-25°05'46"	7.8	V	5.342	0.215	0.348837209	7.536
2018	11	11.1	19 ^h 27 ^m 33 ^s	-25°02'52"	7.8	V	5.342	0.215	0.627906977	7.536
2018	11	11.11	19 ^h 29 ^m 29 ^s	-24°59'52"	7.9	V	5.342	0.215	0.674418605	7.636
2018	11	11.12	19 ^h 31 ^m 25 ^s	-24°56'45"	7.9	V	5.342	0.215	0.720930233	7.636
2018	11	11.13	19 ^h 33 ^m 21 ^s	-24°53'32"	7.9	V	5.342	0.215	0.76744186	7.636
2018	11	11.14	19 ^h 35 ^m 18 ^s	-24°50'13"	7.9	V	5.342	0.215	0.813953488	7.636
2018	11	11.15	19 ^h 37 ^m 14 ^s	-24°46'47"	7.9	V	5.342	0.215	0.860465116	7.636
2018	11	11.16	19 ^h 39 ^m 11 ^s	-24°43'15"	7.9	V	5.342	0.215	0.906976744	7.636
2018	11	11.17	19 ^h 41 ^m 09 ^s	-24°39'36"	7.9	V	5.342	0.215	0.953488372	7.636
2018	11	11.18	19 ^h 43 ^m 06 ^s	-24°35'51"	7.9	V	5.342	0.215	0	7.636
2018	11	11.19	19 ^h 45 ^m 04 ^s	-24°32'00"	7.9	V	5.342	0.215	0.046511628	7.636
2018	11	11.2	19 ^h 47 ^m 02 ^s	-24°28'03"	7.9	V	5.342	0.215	0.093023256	7.636
2018	11	11.21	19 ^h 49 ^m 00 ^s	-24°23'59"	7.9	V	5.342	0.215	0.139534884	7.636
2018	11	11.22	19 ^h 50 ^m 58 ^s	-24°19'49"	7.9	V	5.342	0.215	0.186046512	7.636
2018	11	11.23	19 ^h 52 ^m 57 ^s	-24°15'32"	7.9	V	5.342	0.215	0.23255814	7.636
2018	11	11.24	19 ^h 54 ^m 55 ^s	-24°11'10"	7.9	V	5.342	0.215	0.279069767	7.636
2018	11	11.25	19 ^h 56 ^m 54 ^s	-24°06'41"	7.9	V	5.342	0.215	0.325581395	7.636
2018	11	11.26	19 ^h 58 ^m 53 ^s	-24°02'06"	7.9	V	5.342	0.215	0.372093023	7.636

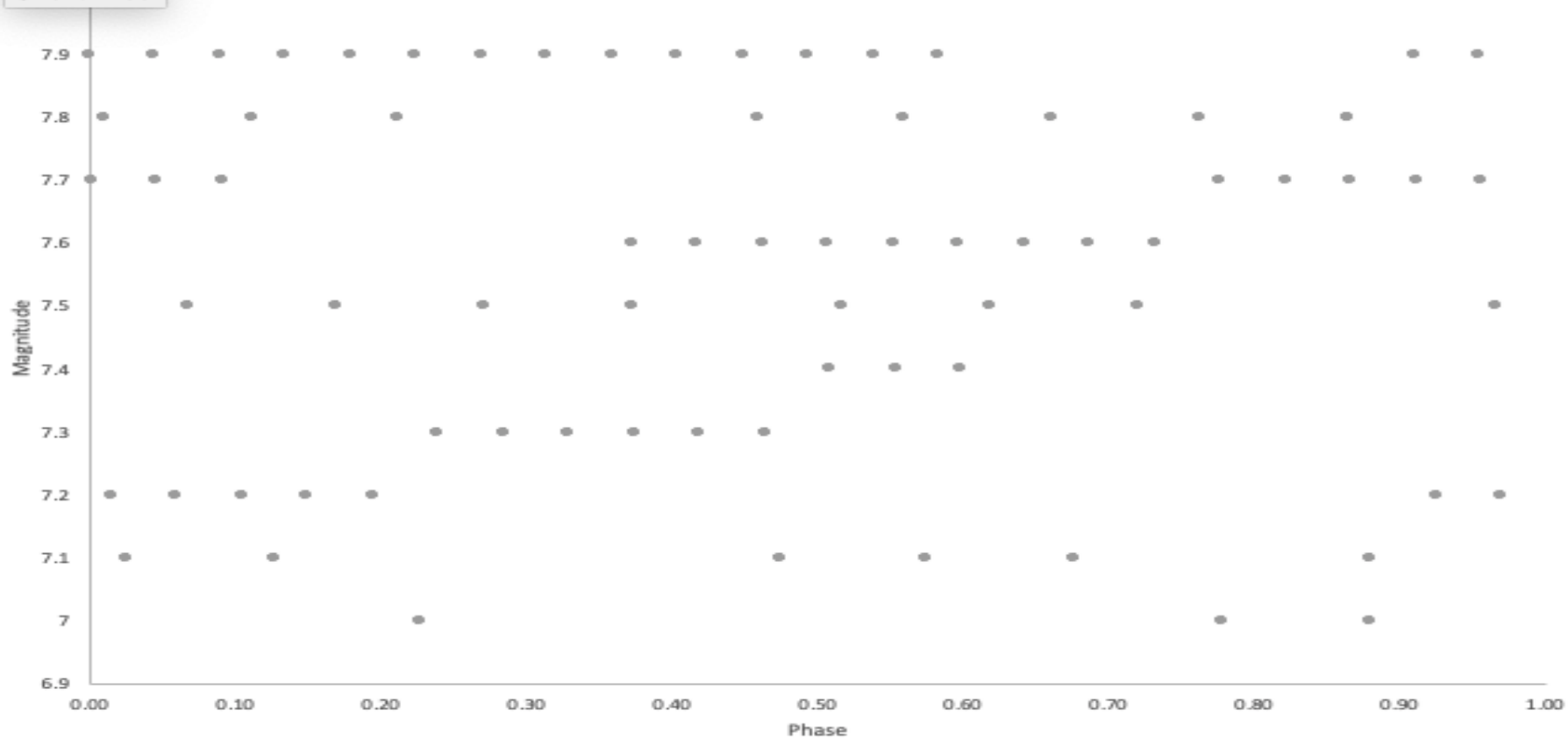
4-Vesta Light Curve



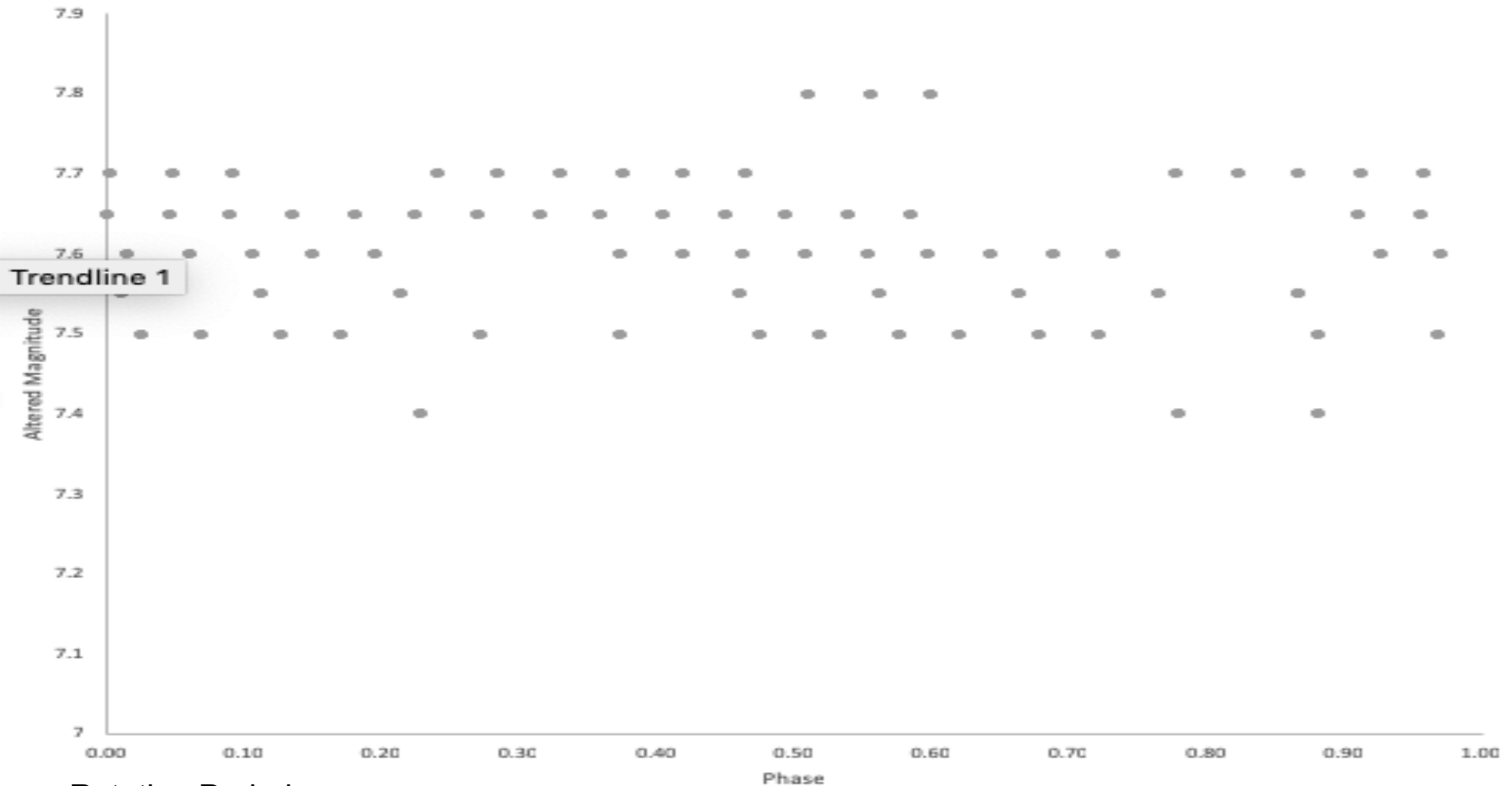


4-Vesta Light Curve

Chart Area



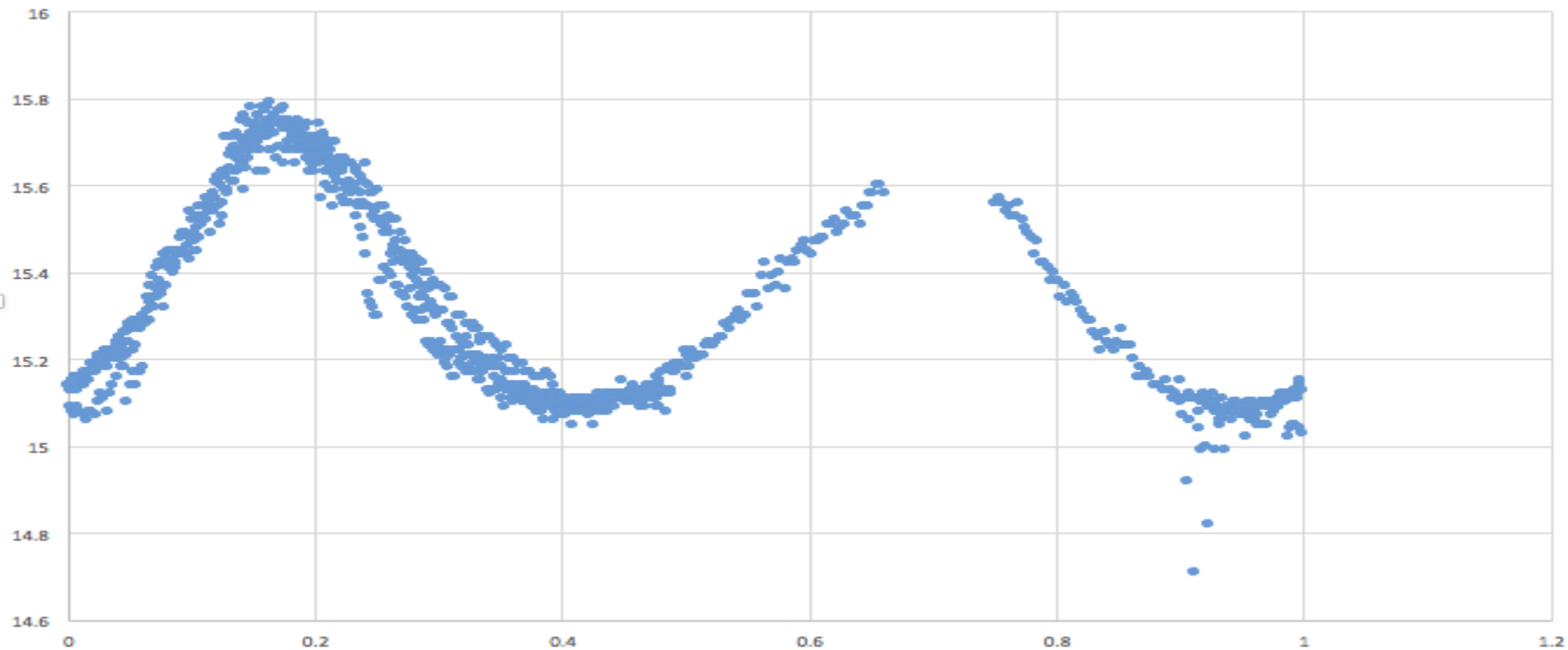
4-Vesta Light Curve



Rotation Period
5.342 hours

What a good Light Curve Looks like

2002 KL6 Lightcurve (altered mag)



Astrophotography - Orion Nebula

- Also known as M42, Orion Nebula is a diffuse nebula situated in the Milky Way, being south of Orion's Belt in the constellation of Orion. It is one of the brightest nebulae, and is visible to the naked eye in the night sky.
- The M42 nebula is estimated to be 24 light years across and its 1,344 ly and has a mass of about 2,000 times that of the Sun.
- The nebula has revealed much about the process of how stars and planetary systems are formed from collapsing clouds of gas and dust.
- Astronomers have directly observed protoplanetary disks, brown dwarfs, intense and turbulent motions of the gas, and the photo-ionizing effects of massive nearby stars in the nebula.
- We initially started with exposures of 1-2 sec, as we used apparent magnitude when calculating exposure time rather than absolute magnitude, which later produced exposure times of around 30 seconds.

Process of stacking/merging images

-Preprocessing: “Raw” Photos

- This was a streamlined process done in astro art made even easier by LCO’s BANZI program, which eliminated the need for any flat fields, dark frames, and dark flats as it already merged them into the best possible picture

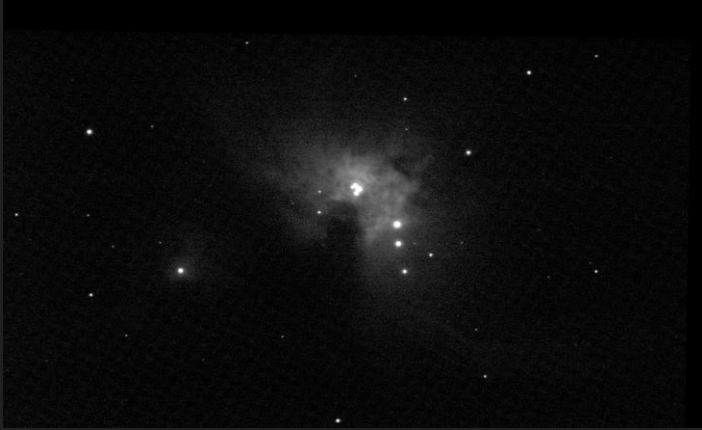
- Mainly used to align, and merge (sigma) multiple captures at once

-Light curves

- This option wasn’t as apparent in Astroart, and it was difficult to dim brighter areas while enlightening darker areas

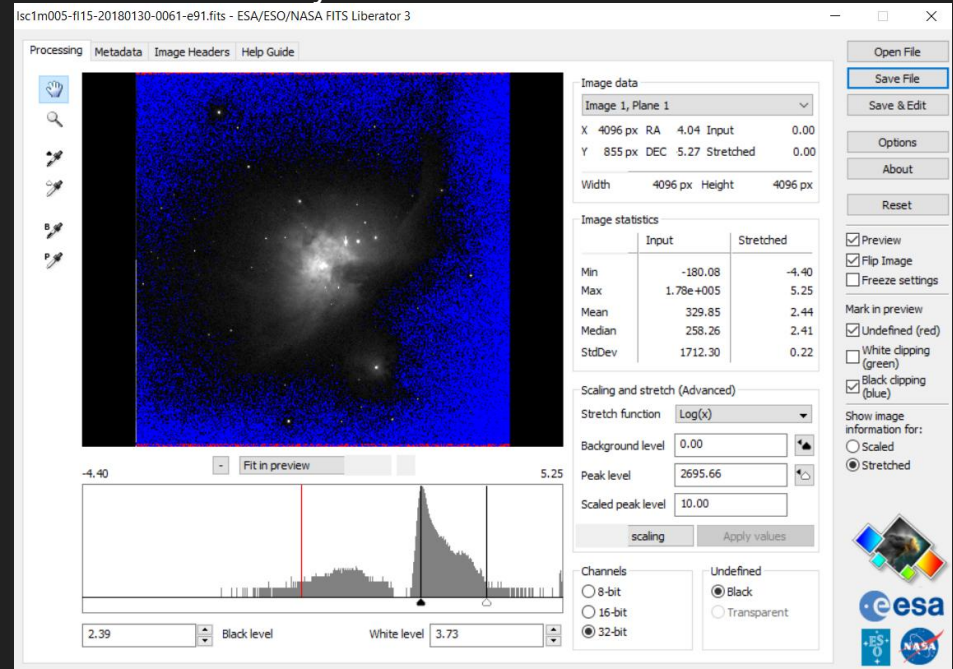
- Then, we used two methods to add color and it was simply a matter of playing with the different options in Astroart/GIMP to produce the best images with the best results

Images after stacking (all of these are through SDSS r' filters)



GIMP

- GIMP is short for GNU Image Manipulation Program, and is a free photoshop tool for Windows, Linux, and Mac
- Before using GIMP, a file conversion from .FITS to .TIFF done by FitsLiberator so GIMP can read the file
- In FitsLiberator, levels could be adjusted and slight brightness control was also available before they were edited further in GIMP
- It has many more options to play around with than Astroart, such as but not limited to:
 - Exposure filters
 - More accurate reading of light levels and curves
 - Saturation and black levels
 - Real time editing previews



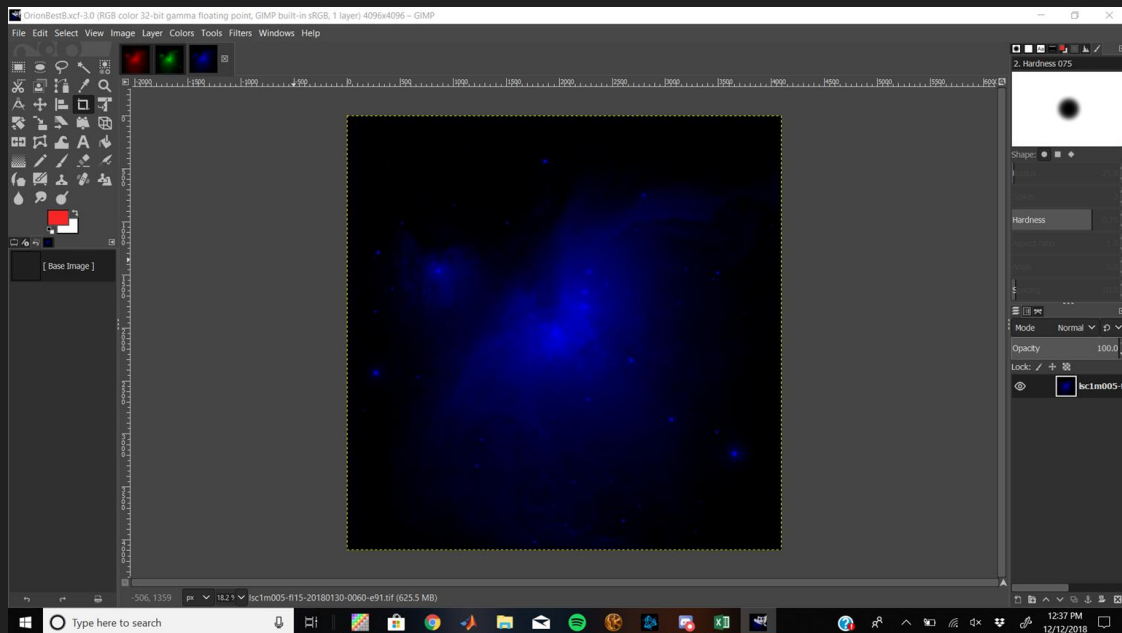
Coloring

-2 methods of coloring

- AstroArt - simple process, as each filter was layered and pre colored in the Trichomy section
- GIMP - a couple more steps involved, but able to play around with this option and see/do more with colors.

-After colors were layered with the correct RGB hues, the light curves and contrast was then able to be adjusted.

-Final touches include contrast levels, hue exposure and color saturation



Merged photos



GIMP images



Andromeda Observation request

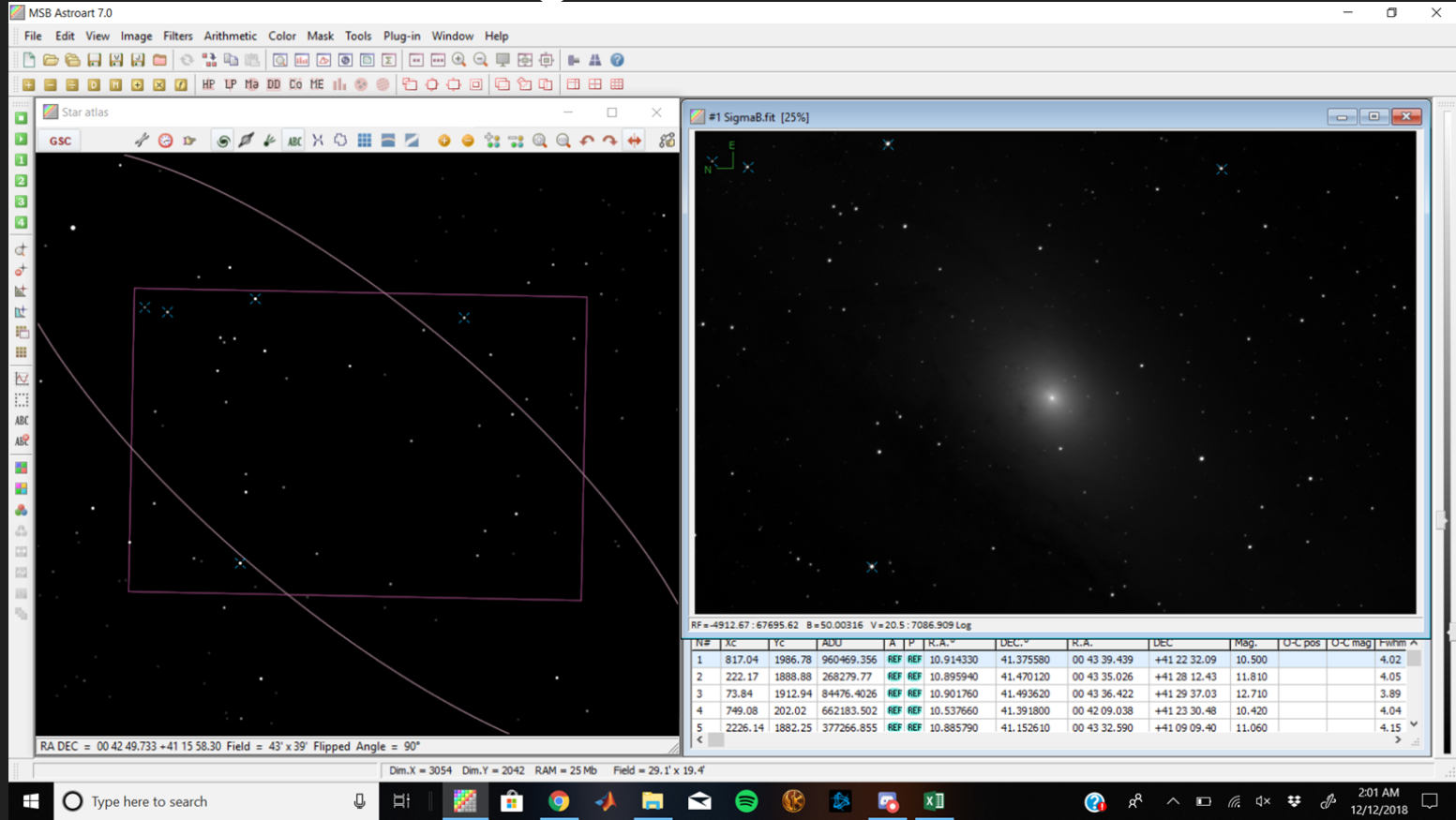
Andromeda Galaxy	0h 42m 44s	+41° 16' 9"	U	3	0.5	11/29-12/05 (UTC) 8:00p.m-10:00p.m 6:00a.m-8:00a.m 3:00a.m-5:00a.m	TFN (Tenerife) OGG (Haleakala) ELP (Fort Davis)
Andromeda Galaxy	0h 42m 44s	+41° 16' 9"	B	3	0.5	11/29-12/05 (UTC) 8:00p.m-10:00p.m 6:00a.m-8:00a.m 3:00a.m-5:00a.m	TFN (Tenerife) OGG (Haleakala) ELP (Fort Davis)
Andromeda Galaxy	0h 42m 44s	+41° 16' 9"	V	3	0.5	11/29-12/05 (UTC) 8:00p.m-10:00p.m 6:00a.m-8:00a.m 3:00a.m-5:00a.m	TFN (Tenerife) OGG (Haleakala) ELP (Fort Davis)
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Andromeda Galaxy	0h 42m 44s	+41° 16' 9"	U	3	1	11/29-12/05 (UTC) 8:00p.m-10:00p.m 6:00a.m-8:00a.m 3:00a.m-5:00a.m	TFN (Tenerife) OGG (Haleakala) ELP (Fort Davis)
Andromeda Galaxy	0h 42m 44s	+41° 16' 9"	B	3	1	11/29-12/05 (UTC) 8:00p.m-10:00p.m 6:00a.m-8:00a.m 3:00a.m-5:00a.m	TFN (Tenerife) OGG (Haleakala) ELP (Fort Davis)
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Andromeda Galaxy	0h 42m 44s	+41° 16' 9"	U	3	2	11/29-12/05 (UTC) 8:00p.m-10:00p.m 6:00a.m-8:00a.m 3:00a.m-5:00a.m	TFN (Tenerife) OGG (Haleakala) ELP (Fort Davis)
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AstroArt w/ Andromeda: Project 3



Andromeda Data: Finding Reference Stars

- Star Atlas

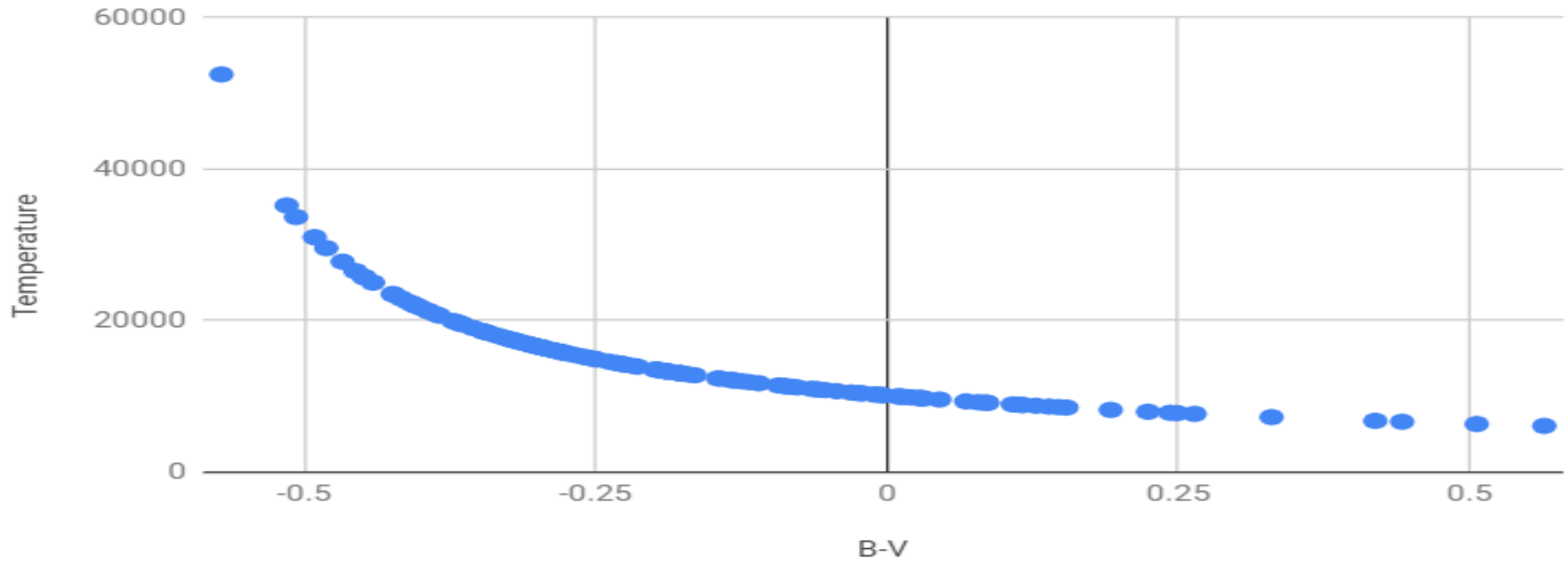


Color Indices to find temperature curve

- Using the star atlas map on astroart we were able to set reference stars
- Using known magnitudes of those reference stars astroart did the calculations for the magnitudes of the rest of the stars in our photo
- We did this for both the B and V filters, then calculated B-V magnitudes of those stars
- Then plugging in those values to this equation we got associated temperature values $((1/(0.92*(B-V)+1.7))+(1/(0.92*(B-V)+0.62)))*4600=T$

Data Analysis: Temperature vs Color Indices

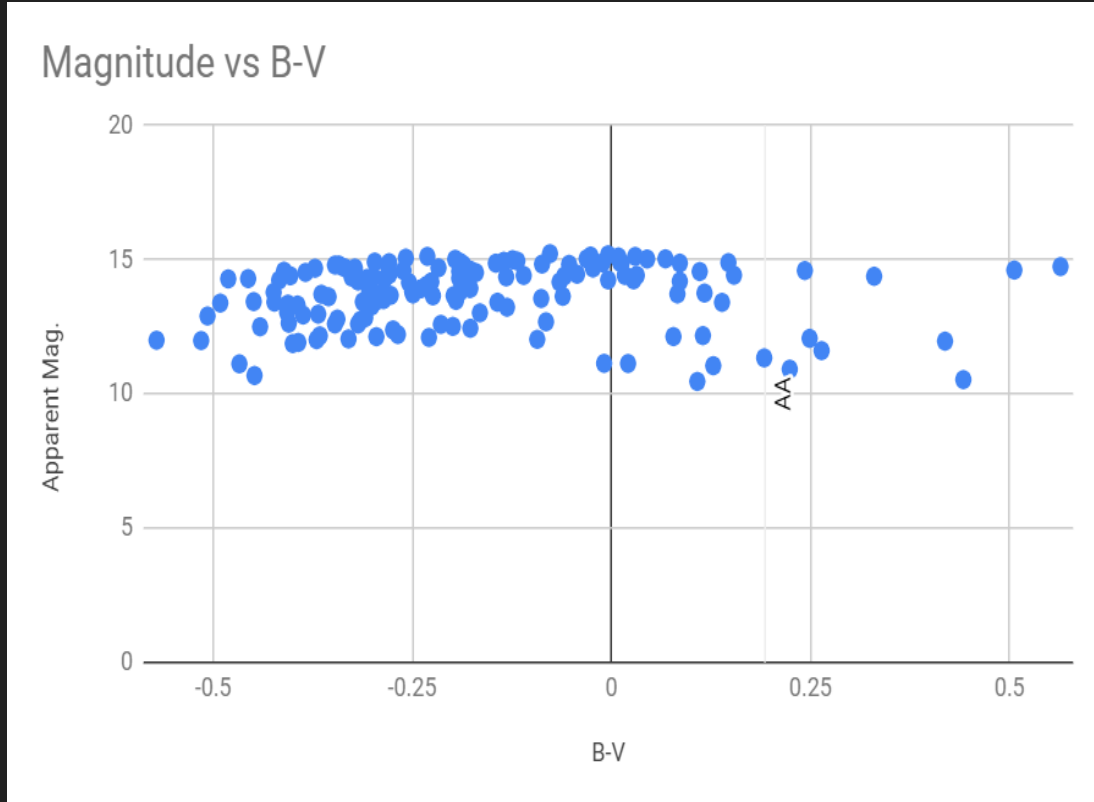
Temperature vs. B-V



Data Analysis

- One of our goals was to construct an H-R diagram for Andromeda
- These are graphs of star Luminosity/Absolute magnitude vs temperature
- We realized we could not calculate luminosity very easily as it depends on the stars distance and radius
- Instead we can procure similar data using known a H-R diagram to compare with our own data
- All of our calculated magnitudes are apparent magnitudes but since the relative scale is the same between both apparent and absolute we can match our data to the known H-R diagram with an absolute magnitude scale
- http://web.physics.ucsb.edu/~jatila/LambdaLabs/Globulars/HRdiagramlab_JKV.pdf

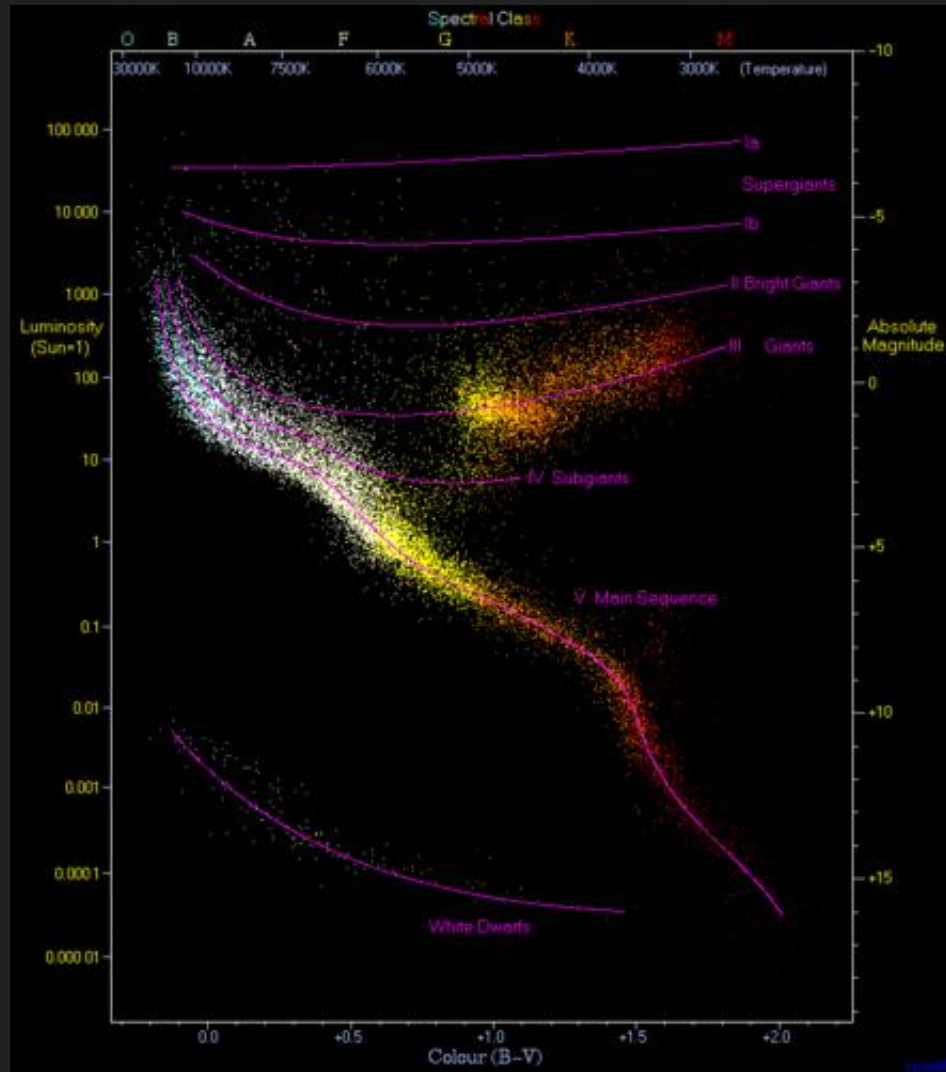
Note: The apparent magnitude should be inverted so that it starts at 20 and goes to 0



The best data points lie in the region of B-V values ranging from -0.5 to 0.25 which have corresponding temperatures in the range of $\sim 40,000\text{K}$ to $\sim 7000\text{K}$

Now we can match this up with the known H-R diagram!

H-R Diagram



Data Analysis: Other methods of distance

- We can't use parallax since Andromeda is too far and the arcsecond that would be too small
- We can use Cepheid Variable stars to measure the distance quite accurately
- A Cepheid variable is a type I; ,star that pulsates radially, varying in both diameter and temperature and producing changes in brightness with a well-defined stable period and amplitude.
- There are Cepheid variable stars located inside Andromeda, as discovered by Edwin Hubble
- By the end of 1924, Hubble had found 36 variable stars in Andromeda, 12 of which were Cepheids.
- This in turn allows one to determine the distance to the star, by comparing its known luminosity to its observed brightness.